

SEATTLE RENTON, WA, USA

WMO: 727934

Lat: 47.493N

Lon: 122.214W

Elev: 9

StdP: 101.22

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 94248

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.3	-1.5	-10.0	1.6	-0.7	-7.0	2.1	1.7	9.1	7.5	8.0	8.3	1.3	100	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.3	30.1	19.1	28.1	18.3	26.5	17.7	20.0	28.3	19.1	26.7	18.2	25.2	3.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.2	21.8	16.2	11.5	21.5	15.3	10.9	21.1	56.9	28.1	53.8	26.8	51.3	25.2	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.6	5.6	-5.7	33.8	2.4	2.1	-7.5	35.4	-8.9	36.6	-10.2	37.8	-12.0	39.4		
(5)			-6.6	21.7	2.4	2.3	-8.3	23.4	-9.7	24.7	-11.1	26.0	-12.9	27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	5.8	6.4	8.3	10.7	14.1	16.9	19.7	19.9	16.8	12.1	8.0	5.4
	DBStd	5.86	3.17	2.76	2.67	2.66	2.91	2.85	2.54	2.29	2.54	2.60	3.25	3.25
	HDD10.0	553	132	102	66	22	2	0	0	0	0	11	74	145
	HDD18.3	2443	387	333	310	230	136	61	13	9	59	195	309	401
	CDD10.0	1299	3	3	13	42	128	207	302	307	204	74	15	2
	CDD18.3	150	0	0	0	0	3	18	56	58	14	1	0	0
	CDH23.3	1308	0	0	2	11	56	187	493	441	118	2	0	0
	CDH26.7	351	0	0	0	1	11	54	145	116	24	0	0	0
Wind	WSAvg	2.4	2.2	2.4	2.6	2.6	2.6	2.6	2.7	2.4	2.2	2.0	2.1	2.2
Precipitation	PrecAvg	1237	172	111	131	103	77	61	25	30	52	119	185	160
	PrecMax	1628	352	243	241	189	136	111	68	115	167	211	416	299
	PrecMin	951	80	0	44	31	21	21	1	4	3	36	69	65
	PrecStd	166	62	52	47	42	28	25	16	26	36	52	77	61
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	15.5	20.0	24.8	28.2	31.1	33.3	32.2	29.4	22.4	16.9	14.0
		MCWB	10.5	9.0	11.3	14.6	16.8	19.2	20.3	20.3	18.1	14.8	13.4	11.3
	2%	DB	12.7	13.1	16.6	20.2	24.7	28.1	30.2	29.5	26.6	19.6	14.9	12.3
		MCWB	10.1	9.0	10.4	12.5	15.9	17.9	19.2	19.2	17.6	14.2	12.1	9.5
	5%	DB	11.4	11.9	14.2	17.6	22.2	25.2	27.9	27.6	24.1	17.7	13.5	11.0
		MCWB	9.1	8.6	9.4	11.4	14.8	16.6	18.4	18.6	16.7	13.4	11.1	8.6
	10%	DB	10.2	10.8	12.6	15.7	19.8	22.7	26.2	25.9	22.3	16.3	12.4	9.8
		MCWB	8.2	8.0	8.8	10.4	13.7	15.5	17.7	17.9	16.0	12.9	10.2	7.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	11.5	12.9	15.3	17.8	20.0	21.4	21.0	19.8	16.9	14.3	12.4
		MCDB	13.6	13.3	17.6	23.2	25.8	29.6	30.4	29.5	26.5	19.4	15.9	13.9
	2%	WB	11.0	10.4	11.6	13.2	16.5	18.4	19.9	20.0	18.3	15.4	12.8	10.4
		MCDB	12.0	12.1	14.8	18.7	23.7	26.5	28.6	28.0	24.5	18.0	14.4	11.5
	5%	WB	9.6	9.2	10.5	12.0	15.2	17.1	18.9	19.1	17.4	14.3	11.7	9.0
		MCDB	10.7	10.9	13.4	16.1	21.2	24.1	27.1	26.3	22.8	16.7	13.0	10.3
	10%	WB	8.4	8.2	9.5	10.9	14.1	16.1	18.0	18.3	16.6	13.4	10.6	7.9
		MCDB	9.7	10.0	12.3	14.4	18.8	22.1	25.3	24.8	21.2	15.8	12.0	9.1
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.7	7.5	8.3	8.9	9.1	10.3	10.3	9.4	7.4	6.2	5.2
		MCDBR	6.6	8.6	10.3	12.5	13.7	14.0	14.3	13.9	13.6	9.9	6.8	5.7
		MCWBR	4.3	5.0	5.6	6.0	6.3	5.9	5.7	5.6	6.0	5.5	4.5	4.1
	5% WB	MCDBR	5.4	6.2	8.5	10.7	12.1	12.5	13.4	12.4	11.6	7.9	5.8	5.1
		MCWBR	4.3	4.2	5.1	5.5	5.8	5.5	5.6	5.3	5.4	5.0	4.3	4.2
Clear-Sky Solar Irradiance	taub	0.328	0.320	0.328	0.363	0.381	0.371	0.356	0.362	0.350	0.333	0.323	0.334	
	taud	2.481	2.538	2.536	2.405	2.377	2.431	2.490	2.482	2.507	2.557	2.534	2.388	
	Ebn at Noon	754	854	900	891	883	893	902	882	862	825	757	684	
	Edh at Noon	66	75	87	109	116	111	103	100	89	73	62	64	
All-Sky Solar Radiation	RadAvg	1.03	1.91	2.87	4.22	5.30	5.85	6.54	5.57	4.04	2.30	1.20	0.83	
	RadStd	0.14	0.25	0.32	0.39	0.43	0.56	0.47	0.37	0.35	0.25	0.16	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (59 neighbors)	N/A	N/A	N/A	N/A	+0.53	N/A	N/A	N/A	+132	+44
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page